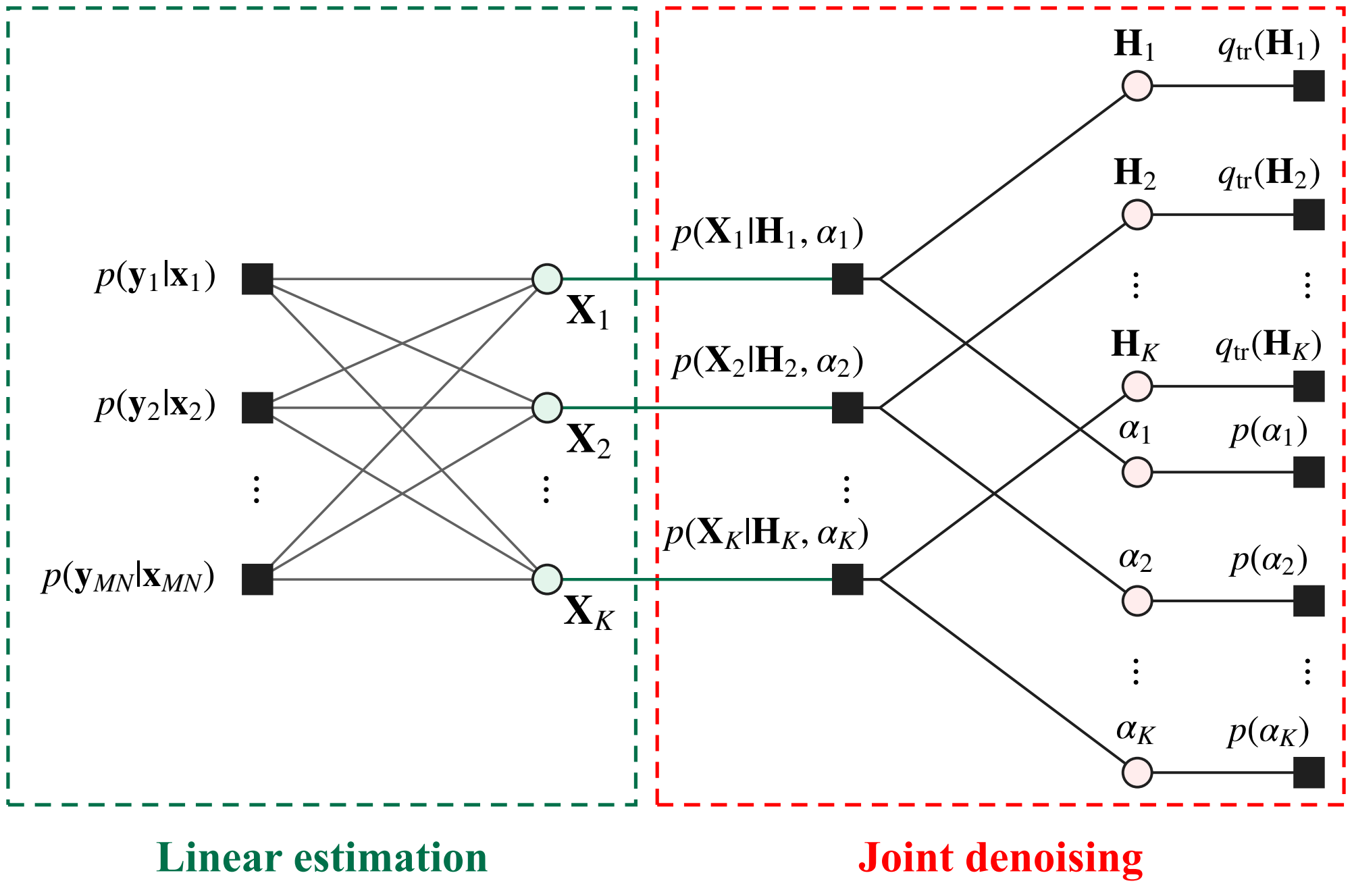
$p(\mathbf{y}_1|\mathbf{x}_1)$
$p(\mathbf{y}_2|\mathbf{x}_2)$
$p(\mathbf{y}_{MN}|\mathbf{x}_{MN})$
$\mathbf{X}_1$
$\mathbf{X}_2$
$\mathbf{X}_K$
$p(\mathbf{X}_1|\mathbf{H}_1, \alpha_1)$
$p(\mathbf{X}_2|\mathbf{H}_2, \alpha_2)$
$p(\mathbf{X}_K|\mathbf{H}_K, \alpha_K)$
$\mathbf{H}_1$
$q_{\text{tr}}(\mathbf{H}_1)$
$\mathbf{H}_2$
$q_{\text{tr}}(\mathbf{H}_2)$
$\mathbf{H}_K$
$q_{\text{tr}}(\mathbf{H}_K)$
α_1
$p(\alpha_1)$
α_2
$p(\alpha_2)$
α_K
$p(\alpha_K)$
Linear estimation
Joint denoising